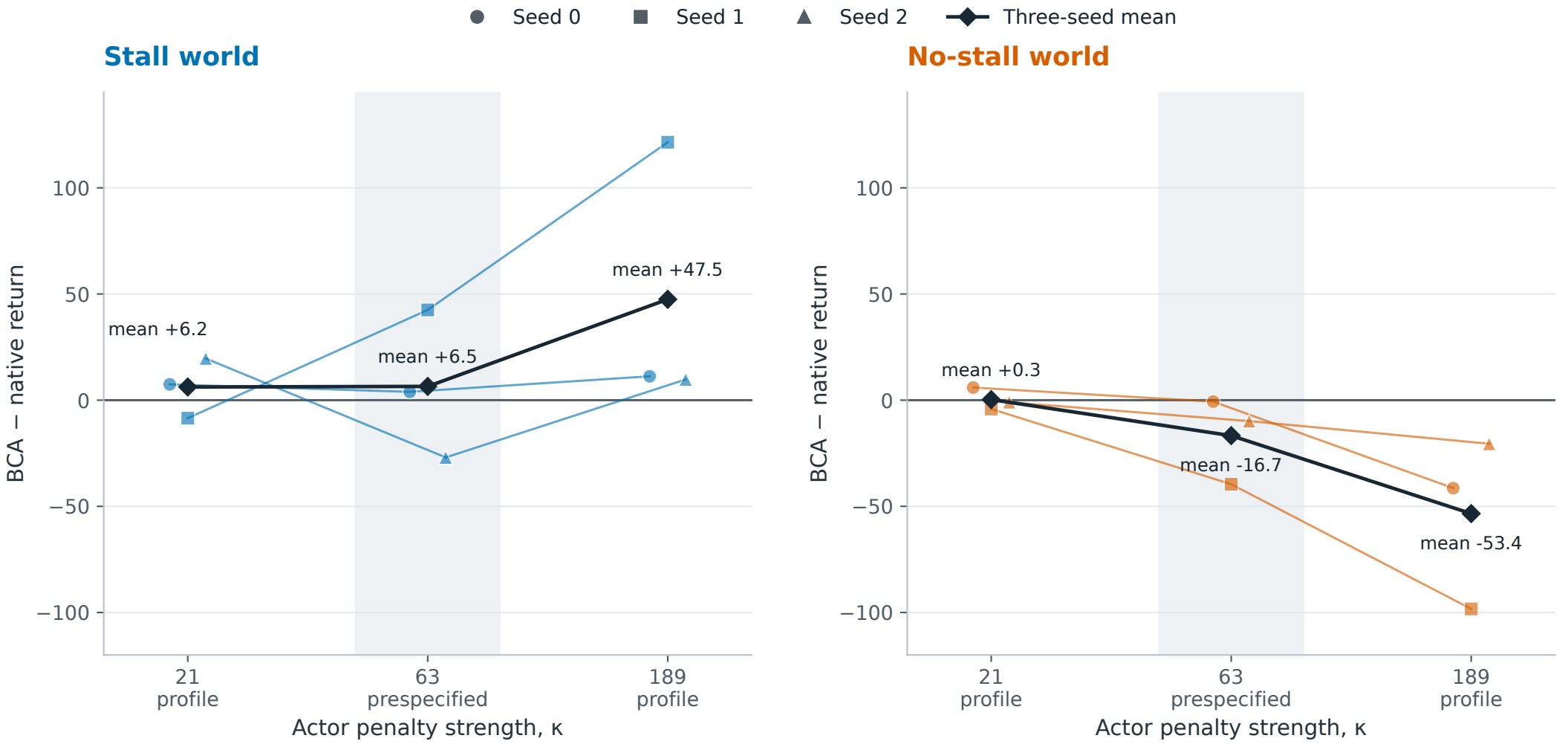
BCA's benefit depends on the world

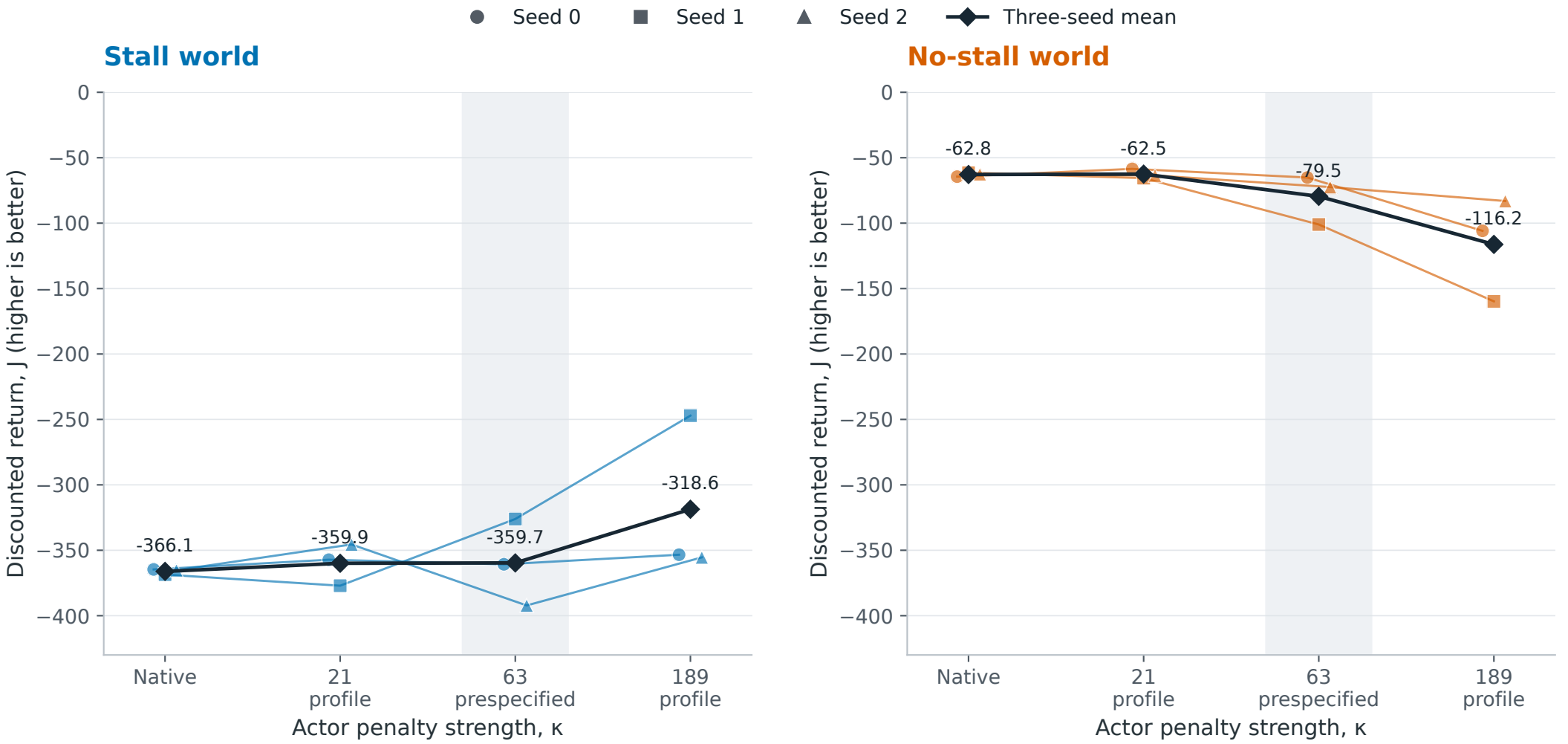
Paired change in discounted policy return relative to native TD3+BC; higher is better.



3 training seeds × 128 common starts per policy. Starts are nested evaluations, not independent training replications.
All strengths shown; κ189 gains in all stall seeds and loses in all no-stall seeds. Formal result remains INCONCLUSIVE (two Qπ checks).

The return scale behind the gains and losses

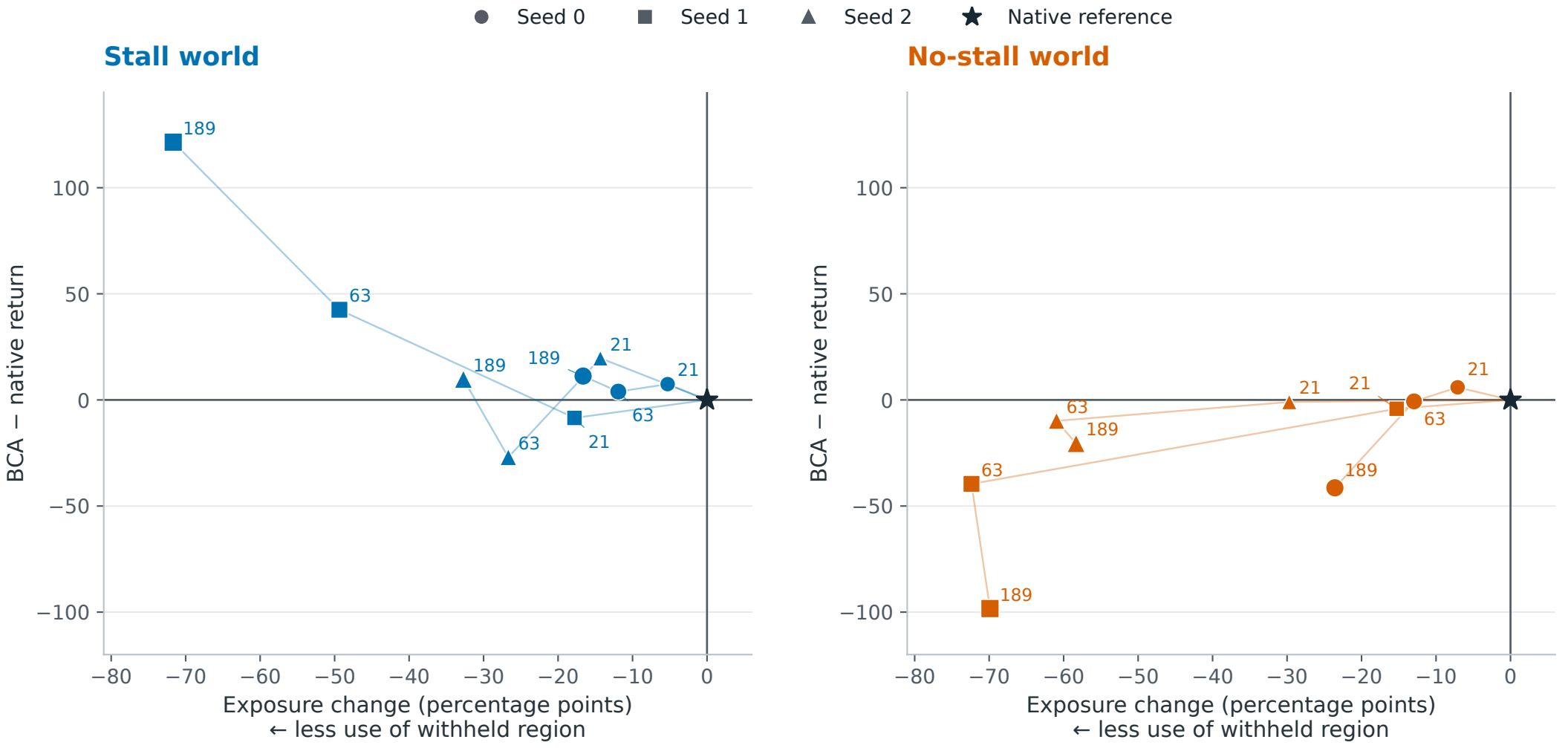
Each point is one final policy's mean across the same 128 starts; a common vertical scale preserves context.



Horizon 100, discount 0.98. BC coefficient unchanged; native and each BCA dose share the same 10k warmup within each seed/world.
All final policies reach 20k cumulative critic updates. Three independent training seeds; no uncertainty interval is inferred from episode counts.

Avoiding unsupported actions can help or hurt

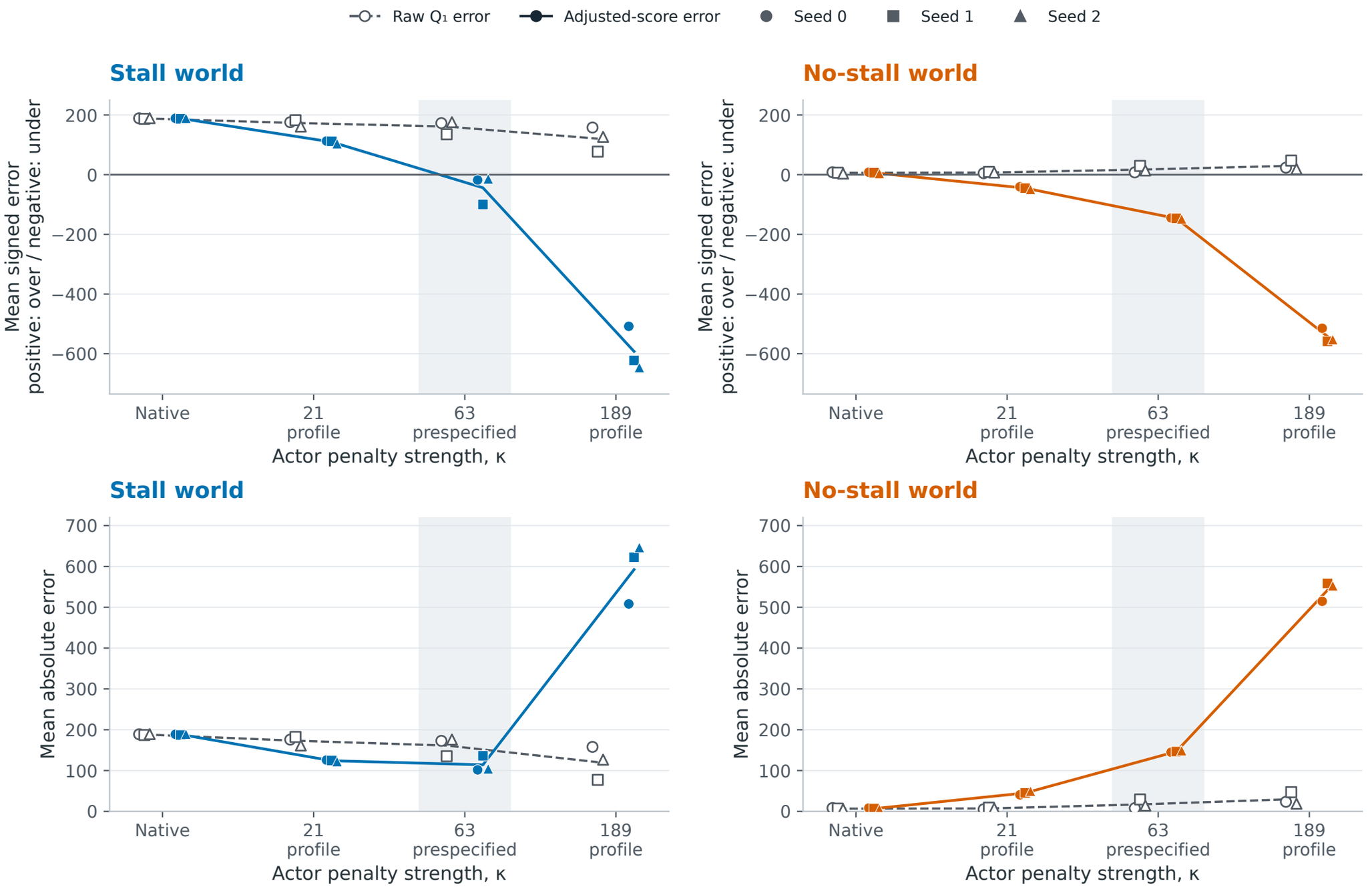
Change in withheld-region exposure versus paired policy-return change. Labels identify κ; shapes identify seeds.



Exposure is the fraction of 128 × 100 executed steps with $|a_0|$ beyond the withheld knee. Each seed/dose is a separate trained policy.
Lines connect the dose profile for readability; they do not establish a causal exposure–return relationship. κ63 was prespecified.

Raw critic optimism and actor-score pessimism

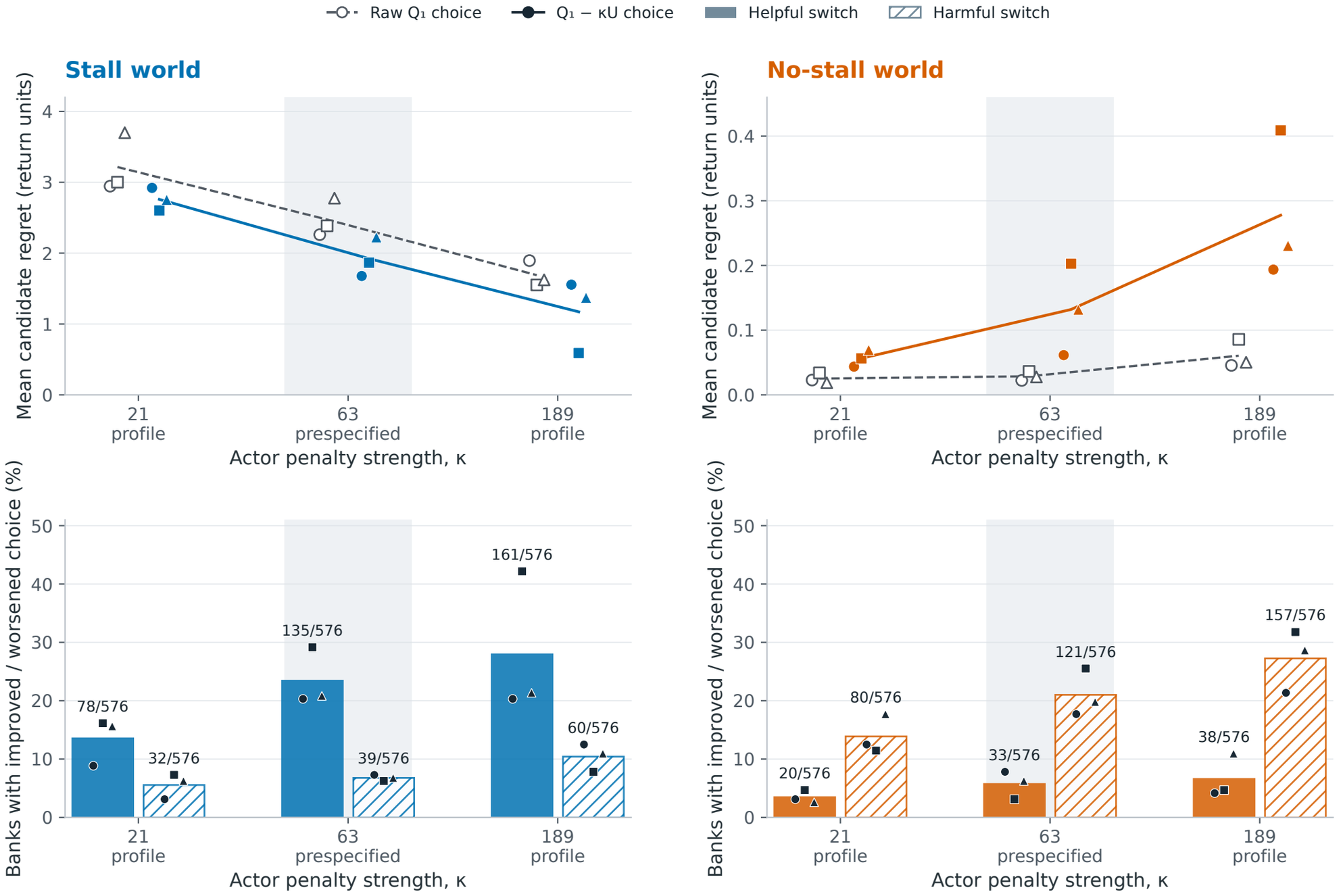
Own-action probes: $Q_1 - Q\pi$ versus $Q_1 - \kappa U - Q\pi$. The actor score is not the learned critic prediction.



48 own-action probes per policy (16 at each of t = 0, 25, 50). Points: training seeds; lines: three-seed means.
Visited states and continuation policies differ across trained policies. These are not common-state critic-accuracy comparisons or true-Q coverage guarantees.

How the penalty changes candidate choices

Retrospective four-candidate banks: logged action plus three withheld a_0 commands; lower regret is better.



192 shared state banks × 3 final policies = 576 evaluations per dose/world, not independent states. Seeds: circle 0, square 1, triangle 2.
Regret uses each policy's own Qπ continuation. This is not deployed actor behavior or proof of the return mechanism. Top-panel scales differ.